ZIYANG CHEN

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Research Interests

Multimodal learning across **vision**, **audio** and **language**; unified multimodal foundation models; diffusion models for multimodal generation.

Work Experience

Research Lead, Luma AI, CA Jan. 2026 – Present

Pre-training: Led audio pretraining and joint audio-video generation across modeling, data, and infrastructure.
Mid- & post-training: Co-led video-action mid-training and led distillation post-training for real-world robot deployment.

Research Scientist, Luma AI, CA June 2025 – Dec. 2025

Post-training: Enabled audio-conditioned generation and distilled video models for efficient inference.

Research Scientist Intern, Adobe Research, CA May 2024 – Nov. 2024

Audio generation: Controllable video-guided Foley generation with diffusion models.

Research Scientist Intern, Meta Reality Labs, PA May 2023 – Nov. 2023

Spatial audio: Real-world acoustic modeling from audio-visual data.

Selected Projects

World Action Model The Open Physical AI Lab — Luma AI

- Co-led video-action model mid-training, developing the model and training infrastructure.
- Led distillation post-training for the world action model, enabling deployment on real-world robots.

Joint Audio-Video Generation Ray3 & **Uni-1** — Luma AI

- Developed an in-house audio variational autoencoder for diffusion-based generation.
- Designed and implemented a unified training framework for multimodal generation.
- Led audio pretraining across speech, sound effects, and music for the Ray3 and Uni model families.
- Designed and implemented joint audio-video pretraining for synchronized audiovisual generation in Ray3 and Uni-1.
- Led the development of large-scale audio and audio-video data curation and annotation pipelines.

Video Avatar Real-Time Speech-to-Video Model — Luma AI

- Designed the model architecture for speech-to-video generation and built the training, inference, and data-filtering pipelines from scratch.
- Applied diffusion distillation to enable avatar interaction driven by a background LLM agent in real-time.

Ray3.14 Flagship Few-Step Video Generation Model — Luma AI

- Core contributor to distillation pipeline and algorithms, enabling fast few-step generation across diverse downstream tasks.

Education

University of Michigan, Ann Arbor Jan. 2022 – May 2025

Ph.D. in Electrical & Computer Engineering | Advisor: Prof. *Andrew Owens*

University of Michigan, Ann Arbor Sep. 2019 – May 2021

M.S.E. in Computer Vision | Advisor: Prof. *Andrew Owens*

Shanghai Jiao Tong University, UM-SJTU JI Sep. 2015 – Aug. 2019

B.S.E. in Electrical & Computer Engineering | Minor in Entrepreneurship
Graduated *magna cum laude*

Publications

* indicates equal contribution

arXiv 2026 — **Luma Uni-1: A Frontier Unified Understanding and Generation Model** Blog
Luma AI · Coming soon

CVPR 2025 — **Video-Guided Foley Sound Generation with Multimodal Controls** Project site
Ziyang Chen, Prem Seetharaman, Bryan Russell, Oriol Nieto, David Bourgin, Andrew Owens and Justin Salamon.

CVPR 2025 — **GPS as a Control Signal for Image Generation** Project site
Chao Feng, Ziyang Chen, Aleksander Holynski, Alexei A. Efros, and Andrew Owens.

CVPR 2025 **Highlight** — **Supervising Sound Localization using In-the-wild Egomotion**
Anna Min, Ziyang Chen, Hang Zhao, and Andrew Owens.

NeurIPS 2024 — **Images that Sound: Composing Images and Sounds on a Single Canvas** Project site
Ziyang Chen, Daniel Geng and Andrew Owens.

CVPR 2024 **Highlight** — **Real Acoustic Fields: An Audio-Visual Room Acoustics Dataset and Benchmark** Project site
Ziyang Chen, Israel D. Gebru, Christian Richardt, Anurag Kumar, William Laney, Andrew Owens, and Alexander Richard.

CVPR 2024 — **Binding Touch to Everything: Learning Unified Multimodal Tactile Representations** Project site
Fengyu Yang*, Chao Feng*, Ziyang Chen*, Hyoungseob Park, Daniel Wang, Yiming Dou, Ziyao Zeng, Xien Chen, Rit Gangopadhyay, Andrew Owens, and Alex Wong.

ICCV 2023 — **Sound Localization from Motion: Jointly Learning Sound Direction and Camera Rotation** Project site
Ziyang Chen, Shengyi Qian and Andrew Owens.

CVPR 2023 — **Conditional Generation of Audio from Video via Foley Analogies** Project site
Yuexi Du, Ziyang Chen, Justin Salamon, Bryan Russell, and Andrew Owens.

CVPR 2023 **Highlight** — **Self-Supervised Video Forensics by Audio-Visual Anomaly Detection** Project site
Chao Feng, Ziyang Chen and Andrew Owens.

ECCV 2022 — **Sound Localization by Self-Supervised Time Delay Estimation** Project site
Ziyang Chen, David F. Fouhey and Andrew Owens.

CVPR 2022 — **Mix and Localize: Localizing Sound Sources in Mixtures** Project site
Xixi Hu*, Ziyang Chen* and Andrew Owens.

CoRL 2021 **Oral** — **Structure from Silence: Learning Scene Structure from Ambient Sound** Project site
Ziyang Chen*, Xixi Hu* and Andrew Owens.

Academic Service

Workshop Organizer

Generative AI for Audio-Visual Content Creation — ICCV 2025 · ECCV 2026

Sight and Sound — CVPR 2024 · CVPR 2025 · CVPR 2026

Reviewer

Conferences — CVPR (2023–2026), ICCV (2023, 2025), ECCV (2024), NeurIPS (2024), SIGGRAPH (2024), WACV (2023)

Journals — IJCV

Awards & Scholarships

Lums Fellowship, University of Michigan, Ann Arbor — 2019–2021

Outstanding Graduate — Class of 2019, Shanghai Jiao Tong University — 2019

Academic Excellence Scholarship, Shanghai Jiao Tong University — 2016–2018

Dean's List, Shanghai Jiao Tong University, SJTU–UM JI — 2015–2018